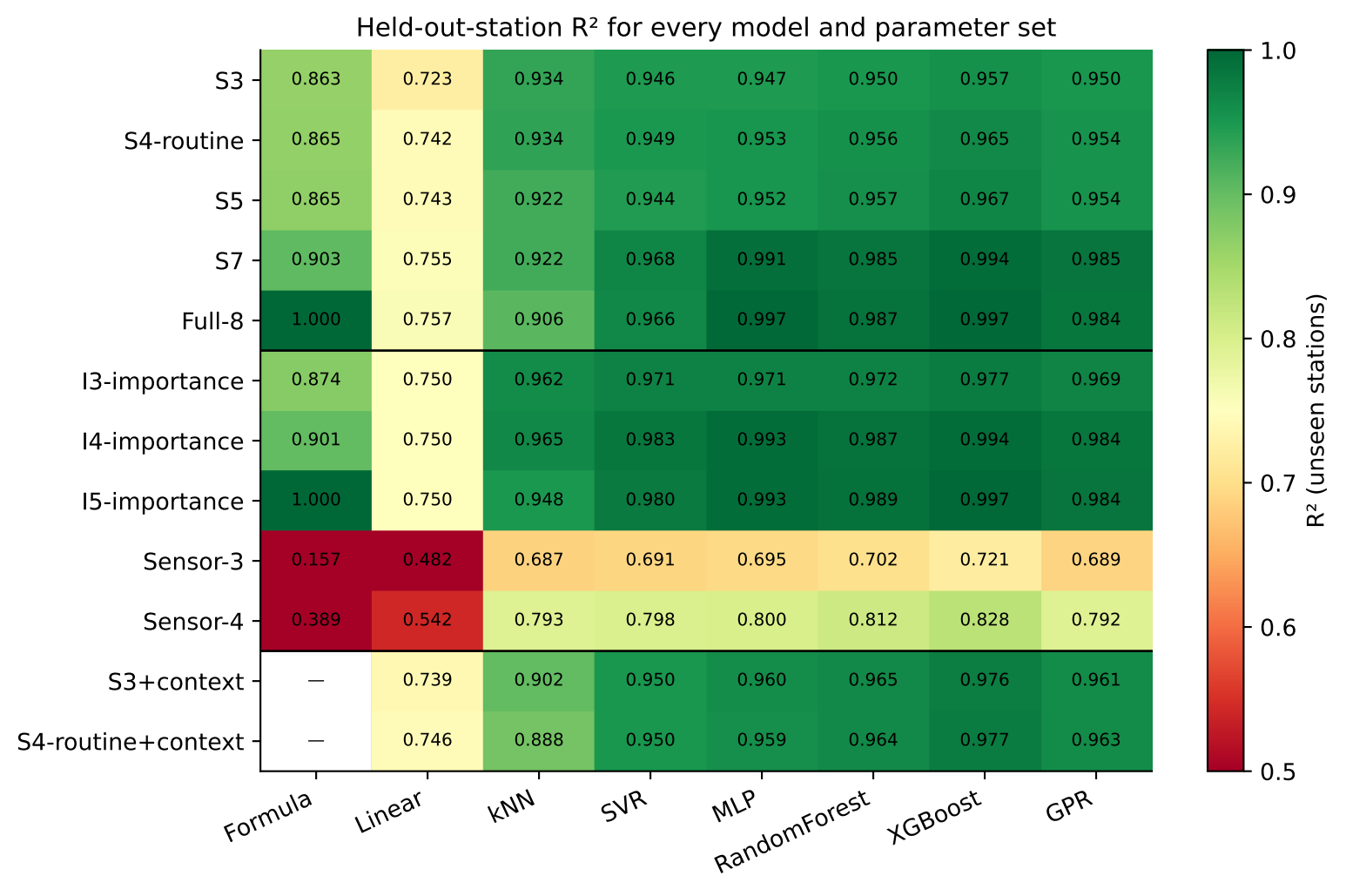
Held-out-station R² for every model and parameter set
S3
S4-routine
S5
S7
Full-8
I3-importance
I4-importance
I5-importance
Sensor-3
Sensor-4
S3+context
S4-routine+context
0.863 0.723 0.934 0.946 0.947 0.950 0.957 0.950
0.865 0.742 0.934 0.949 0.953 0.956 0.965 0.954
0.865 0.743 0.922 0.944 0.952 0.957 0.967 0.954
0.903 0.755 0.922 0.968 0.991 0.985 0.994 0.985
1.000 0.757 0.906 0.966 0.997 0.987 0.997 0.984
0.874 0.750 0.962 0.971 0.971 0.972 0.977 0.969
0.901 0.750 0.965 0.983 0.993 0.987 0.994 0.984
1.000 0.750 0.948 0.980 0.993 0.989 0.997 0.984
0.157 0.482 0.687 0.691 0.695 0.702 0.721 0.689
0.389 0.542 0.793 0.798 0.800 0.812 0.828 0.792
— 0.739 0.902 0.950 0.960 0.965 0.976 0.961
— 0.746 0.888 0.950 0.959 0.964 0.977 0.963
Formula
Linear
kNN
SVR
MLP
RandomForest
XGBoost
GPR
R² (unseen stations)
1.0
0.9
0.8
0.7
0.6
0.5